

Machine Learning for Attack Detection in IoT Networks

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Abstract—IoT has critical data stream sources that produce continuously massive amounts of data from numerous applications. It is, then, essential to ensure that the data streams are safe, secure, and protected from cyberattacks. Indeed, cyberattacks have evolved with the emerging IoT technologies and created challenges to the IoT system's confidentiality, integrity, and availability. Hence, Intrusion detection systems could give a vigorous layer of defence in the IoT systems by controlling and monitoring events in the network to mitigate intrusion and identify anomalies in a data stream. For this purpose, we studied two recommended datasets for anomaly detection in IoT network systems: Bot IoT and TON IoT datasets. We studied the security aspect of the IoT network by investigating the machine learning schema usability in anomaly detection found within the IoT network data. The considered machine learning models are: Naive Bayes (NB), SVM with Linear, RBF, Sigmoid and polynomial kernels, and Multi-Layer Perceptron (MLP). The studied performance metrics are: accuracy, detection rate, recall, precision, and F1-score. The results show that the NB, Linear SVM, RBF SVM, polynomial SVM, and MLP models provide interesting result with binary classification. However, the Linear SVM and MLP model outperform the other models with the multi-class classification.

Index Terms—Internet of Things (IoT), Machine Learning (ML), Intrusion Detection (ID), IoT datasets.

I. INTRODUCTION

Cyber-security is a challenging topic specifically when emerging technologies, like the Internet of Things (IoT), are involved [1]. The IoT is expanding in many application domains, such as industries, smart homes, and smart cities [2]. The IoT proliferation causes important risks to data confidentiality, integrity, and availability [3]. Indeed, malicious users could exploit applications and network traffic vulnerabilities to threaten the IoT systems [4]. It was noted that IoT security draws attention to different researchers and developers as many IoT services are deploying and increased cyber-attacks are occurring. Yet, the need to protect IoT systems from attacks is crucial [5].

It has been shown that the Intrusion Detection System (IDS) is an important defence line technique, which is deployed, often, with other security techniques such as encryption and access control [6]. The IDS implemented for IoT systems could face the challenge of attacks or malicious patterns recognition. Since Machine Learning (ML) tools have proven their performances in anomaly detection, IDS could leverage these tools to solve security issues [6], [7], [8], [9], [10], [11].

Thus, the selection of a suitable method for ML for the IoT system in addition to a good dataset choice with realistic IoT traffic for training can help in intrusion detection. However, there have been limited investigations whose purpose is to detect a variety of attacks that compromise the CIA triad using unbalanced datasets. The unbalanced dataset is a realistic case of study since some attacks are more frequent than others. As a result, detecting the less frequent attacks is a challenge to address.

Therefore, our study highlights, via different datasets, the intrusion detection performance of a variety of high-risk attacks using machine learning.

This work investigates different models for intrusion detection in IoT networks using ML methods. In summary, the keys contributions of this paper are as follows:

- We have studied two relevant datasets for IDS anomaly detection for IoT networks (i.e., BoT-IoT and TON-IoT).
- We have proposed different models for IDS anomaly detection using a bunch of machine learning methods. In addition, we studied the performance of these models using accuracy, detection rate, false alarm, recall, precision, and F1-score performance metrics.

The remainder of this paper is organized as follows: Section II discusses recent related works. Section III explores the different studied datasets. Then, discusses the proposed models by studying the data preprocessing process, the different exploited ML methods, and the evaluation results. Finally, section IV presents a summary of this paper and provides the perspectives of this work.

II. RELATED WORKS

ML is a powerful tool for developing anomaly detection systems for IoT networks. In this section, we investigate different recent works-based ML methods to detect network intrusion in IoT systems. These methods are categorized as supervised ML and unsupervised ML methods. We notice that supervised ML methods are widely used for intrusion detection in IoT networks compared to the unsupervised ML methods. Indeed, it provides a better accuracy and efficiency. In this section, we describe the contributions of the different recent related works that investigate ML models to detect anomalies in IoT networks.

A.Sarkar-&al. [12] investigate Ensemble learning methods for detecting DoS, probe, R2L, U2R attacks. The exploited datasets are: KDD, Cup99 and NSL-KDD. Accuracy was the studied performance evaluation metric. The authors propose to increase the weight of the best algorithms predictions to boost the detection accuracy. However, the work should investigate IoT network realistic traffic and should Investigate ML methods more thoroughly.

T. Khoei-&al. [13] investigate Gaussian Naïve Bayes, Classification and Regression Decision Tree, Logistic Regression, C-Support Vector Machine, Light Gradient Boosting, Alex Neural Network, Principal Component Analysis, K-means, and Variational Autoencoder methods. CICDDOS 2019 was the exploited dataset. The studied performance metrics are: accuracy, probability of detection, probability of misdetection, probability of false alarm, processing time, prediction time, training time per sample, and memory size. This research work found that the Alex Neural Network model outperforms the other supervised models. The Variational Autoencoder model has the best results compared to unsupervised models. However, this work should use different datasets with realistic attacks for validation.

N. Saran-&al. [14] investigate K-Nearest Neighbour (K-NN), Support Vector Machine (SVM, Naïve Byes (NB), Random Forest (RF), Decision Tree(DT), and Stochastic Gradient Descent(SGD) methods for detecting Aggressive Scan (A-Scan), UDP Scan (U-Scab), Sparta Secure Shell (SSH), brute-force, and MQTT brute-force attacks. MQTT-IoT-IDS2020 is the exploited dataset. The studied performance metrics are : Precision, Recall, and F1-Score. The experiment results derive more than 97% overall accuracy for all implemented ML classifier algorithms. However, this work should use different datasets with realistic attacks.

M.ElKashlan-&al. [15] investigate Naïve Bayes (NB), J48 classifier, the attribute-select classifier, and the Filtered classifier methods for detecting command-and-control, DDoS, Okiru, and Part of a horizontal port scan. IoT-23 was the exploited dataset. the studied performance metrics are: Precision, Accuracy, Recall, and F1-Score. This research work found that the filtered classifier is the best model for anomaly detection and classification using the IoT23 dataset for IoT Electric Vehicle Charging Stations (EVCSSs). Indeed, it achieves the highest accuracy in both binary classification and multiclass classification. However, this work should study the impact of feature selection.

M.Mohy-eddine-&al. [16] investigate the K-Nearest Neighbors (K-NN) method and exploit the Bot-IoT dataset. the studied performance metrics are: Accuracy, Detection rate, and False alarm rate. This research work exploit the principal component analysis (PCA) and the genetic algorithm (GA) as feature selection methods to select the ten best performing features. The model provided 99.99% accuracy. However, this work consider only binary classification, it should investigate also the multi-class classification.

K. Sasikala-&al. [17] investigate the logistic regression method for detecting Backdoor, Fuzzers, Reconnaissance, Ex-

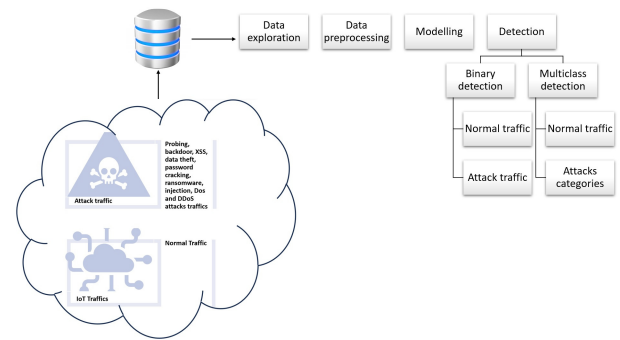


Fig. 1: The proposed approach

ploits, Analysis, DoS, Worms, and Generic attacks. NB15 dataset was the exploited dataset. Accuracy is the studied performance metric. The results show anomalies predictions with a 98% accuracy using the automated feature selection approach. The logistic regression accuracy using binary classification is 97.80%, and it is equal to 97.58% using multi-class classification. However, this work should study other performance metrics.

III. PROPOSED MODELS AND RESULTS

ML approaches provide interesting tools to detect new attacks and thus, prevent unauthorized access. In this section, we present the different implemented models based on ML, which are developed based on the Cross-Industry Standard Process (CRISP) [18]. Fig 1 describes our proposed approach. It is pursuing the typical IDS components: data source, pre-processing, decision core, and response modules. The different phases of our proposed models are described in this section.

A. Data exploration

To identify and respond to attacks automatically, ML could be leveraged for this purpose. They can improve the detection rate of traditional static signature-based detection system. Accordingly, we need to exploit the available public datasets to train and test ML models. Darpa98 [19], KDD99 [20] and NSL KDD [21] [22] are well-known datasets in intrusion detection studies, which were generated a decade ago. They were studied and leveraged for many intrusion detection systems research works. However, these datasets do not contain IoT features in their recorded traffic. Therefore, we have targeted to investigate datasets that accurately mimic IoT network traffic activities, which are : BoT-IoT [23], [24], [25] and ToN-IoT [26], [27], [28], [29], [26].

The two datasets have simulated different realistic attack scenarios in the IoT environment. A comparison between the different attack scenarios is presented in Table I. Some attacks have the same purpose but they have been labelled differently in the different datasets. Therefore, in Table I we indicate the attack and how it was achieved (or labelled) in a dataset. Tables II and III illustrate the sample records for different traffic categories in BoT-IoT and TON-IoT datasets of our study. It is clear that the datasets are unbalanced.

The attacks	BoT-IoT	TON-IoT
DoS	Yes	Yes
DDoS	Yes	Yes
Probing	Yes: it is achieved through Reconnaissance attack	Yes: it is achieved through scanning attack
Injection attack	No	Yes
Data theft	Yes	No
Backdoor	No	Yes
Ransomware	No	Yes
XSS	No	Yes
Password cracking attack	No	Yes

TABLE I: Different simulated attacks in the datasets

Category	Instances No.	Instances %
Normal	477	0,013%
Dos/DDos	3576884	97,502%
Reconnaissance	91082	2,483%
Information Theft	79	0,002%
Total	3668522	100%

TABLE II: BoT-IoT traffic category records

B. Data preprocessing

The data preprocessing phase seeks to simplify subsequent treatments by using common strategies such as correction and normalization. It aims at keeping the relevant data, and converts all non-numerical values into numerical ones.

The preparing Dataset phase focuses on evaluating and analysing the features to be selected from the dataset. Features selection processes by selecting relevant features subset from the original dataset features for use in the model construct. In this work, we used the correlation coefficient method, which is a filter technique for feature selection [18].

C. Modelling

The different models were created based on ML algorithms. We proposed models using ML algorithms, which are Naive Byes (NB) [30], [31], [32], [33], [34], Support Vector Machine (SVM) with different kernels (linear, polynomial, radial basis function) [35], [36], and Multi Layer Perceptron (MLP) [37].

D. Evaluation

This section discusses the performance of different ML models (i.e, NB, Linear SVM, RBF SVM, Sigmoid SVM, Polynomial SVM, and MLP) for intrusion detection purpose using BoT-IoT and TON-IoT datasets. Obviously, the TON-IoT datasets are composed of the Fridge, Garage door, GPS, Sense motion, Modbus, Thermostat, and Weather datasets.

	Fridge		GPS		Sense motion		Garage door		Modbus		Thermostat		Weather	
Category	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
Normal	35000	63%	35000	59%	35000	56%	35000	59%	35000	68%	35000	66%	35000	59%
Backdoor	2902	5%	5000	8%	5000	8%	5000	8%	5000	10%	5000	9%	5000	8%
DDoS	5000	9%	5000	8%	5000	8%	5000	8%	-	-	-	-	5000	8%
Injection	2902	5%	5000	8%	5000	8%	5000	8%	5000	10%	5000	9%	5000	8%
Password-cracking	5000	9%	5000	8%	5000	8%	5000	8%	5000	10%	5000	9%	5000	8%
Ransomware	2902	5%	2833	5%	5000	8%	2902	5%	-	-	2264	4%	2865	5%
Scanning			550	1%	1775	3%	529	1%	529	1%	61	0%	529	1%
XSS	2042	4%	577	1%	449	1%	1156	2%	577	1%	449	1%	866	1%
Total	55748	100%	58960	100%	62224	100%	59587	100%	51106	100%	52774	100%	59260	100%

TABLE III: TON-IoT traffic category records

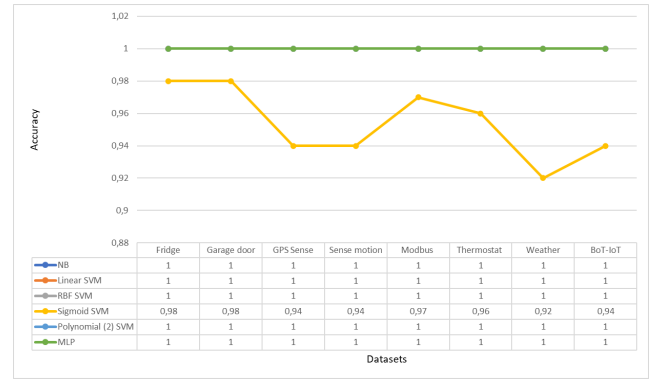


Fig. 2: Accuracy performance metric for the binary classification

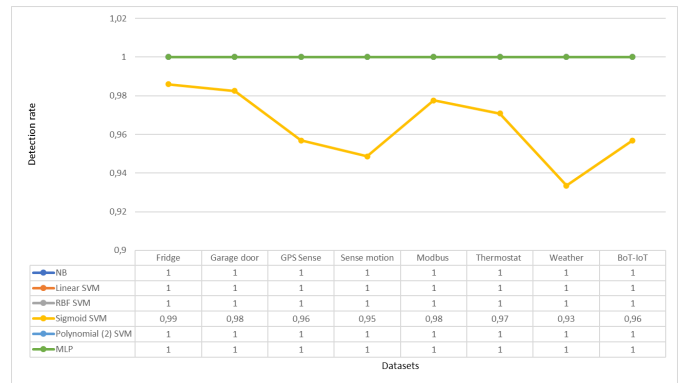


Fig. 3: Detection rate performance metric for the binary classification

Hence, we can say that our models were performed on eight datasets.

The two datasets (i.e., BoT-IoT and TON-IoT) were split into train sets (80%) and test sets (20%). In addition, they were used to conduct binary and multiclass classification experiments using the proposed anomaly detection models.

The different performance metrics, used to evaluate the performance of detection, are : accuracy, Detection Rate (DR), Recall (Re.), Precision (Pre.), F1-Score (F1.) [38], [39].

1) *Binary classification results:* Our results indicate that the NB, Linear SVM, RBF SVM, polynomial SVM, and the MLP models show interesting results with an overall score of 1.00 for the different performance metrics. However, the sigmoid SVM model shows the lowest score for accuracy, detection rate, precision, recall, and F1-score.

Figure 2 shows the accuracy performance metric with the binary classification for the eight studied datasets.

Figure 3 exposes the average detection rate performance metric with the binary classification for the eight datasets.

Table IV demonstrates Precision, recall and F1-score performance metrics with the binary classification for the BoT-IoT dataset.

Table V shows Precision, recall and F1-score performance metrics with the binary classification for TON-IoT datasets.

Models	Traffic	Pre.	Re.	F1.
NB	Normal	1	1	1
	Attack	1	1	1
Linear SVM	Normal	1	1	1
	Attack	1	1	1
RBF SVM	Normal	1	1	1
	Attack	1	1	1
Sigmoid SVM	Normal	0.95	0.96	0.95
	Attack	0.94	0.92	0.93
Polynomial SVM	Normal	1	1	1
	Attack	1	1	1
MLP	Normal	1	1	1
	Attack	1	1	1

TABLE IV: Precision, recall and F1-score performance metrics with the binary classification for the BoT-IoT dataset

Class	NB			Linear SVM			RBF SVM			Sigmoid SVM			Polynomial SVM			MLP		
	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.
Normal	1	1	1	1	1	1	1	1	1	0.99	0.99	0.99	1	1	1	1	1	1
Attack	1	1	1	1	1	1	1	1	1	0.98	0.98	0.98	1	1	1	1	1	1

(a) Performance metrics with binary classification for the Fridge dataset

Normal	1	1	1	1	1	1	1	1	1	0.98	0.98	0.98	1	1	1	1	1	1
Attack	1	1	1	1	1	1	1	1	1	0.97	0.97	0.97	1	1	1	1	1	1

(b) Performance metrics with binary classification for the Garage Door dataset

Normal	1	1	1	1	1	1	1	1	1	0.95	0.96	0.95	1	1	1	1	1	1
Attack	1	1	1	1	1	1	1	1	1	0.94	0.92	0.93	1	1	1	1	1	1

(c) Performance metrics with binary classification for the GPS dataset

Normal	1	1	1	1	1	1	1	1	1	0.95	0.95	0.95	1	1	1	1	1	1
Attack	1	1	1	1	1	1	1	1	1	0.93	0.93	0.93	1	1	1	1	1	1

(d) Performance metrics with binary classification for the Sense Motion dataset

Normal	1	1	1	1	1	1	1	1	1	0.98	0.98	0.98	1	1	1	1	1	1
Attack	1	1	1	1	1	1	1	1	1	0.95	0.95	0.95	1	1	1	1	1	1

(e) Performance metrics with binary classification for the Modbus dataset

Normal	1	1	1	1	1	1	1	1	1	0.98	0.97	0.97	1	1	1	1	1	1
Attack	1	1	1	1	1	1	1	1	1	0.94	0.95	0.95	1	1	1	1	1	1

(f) Performance metrics with binary classification for the Thermostat dataset

Normal	1	1	1	1	1	1	1	1	1	0.93	0.93	0.93	1	1	1	1	1	1
Attack	1	1	1	1	1	1	1	1	1	0.9	0.9	0.9	1	1	1	1	1	1

(g) Performance metrics with binary classification for the Weather dataset

TABLE V: Performance metrics with the binary classification for the TON-IoT datasets

2) *Multiclass classification results:* Our results reveal that the Linear SVM and the MLP show the better performance metrics for the different studied datasets. However, the Sigmoid SVM shows the worst performance metrics and ineffectiveness for low samples detection. The Sense motion dataset provides the lowest accuracy and detection rate performances for the different models. Indeed, the relevance of features in a dataset is important for the detection performance. However, the machine learning models take as inputs two types of features, which are temporal (Timestamp) and physical (motion and light status). These features remain irrelevant and insufficient for attack detection in the Sense motion dataset.

Figure 4 displays the accuracy performance metric with the multi-class classifications for the eight datasets.

Figure 5 reveals the average detection rate performance metric with the multi-class classification for the eight datasets.

Table VI displays the precision, recall, and F1-score with the multiclass classification for the BoT-IoT dataset.

The table VII exposes the precision, recall, and F1-score performance metrics with the multi-class classification for the TON-IoT datasets. For some datasets in the TON IoT datasets, some classes were not considered. Therefore, we used the symbol (-) in the result of these unconsidered classes in the table VII.

3) *Discussion:* We studied unbalanced datasets for IoT networks: the BoT-IoT and the TON-IoT datasets. Then, we

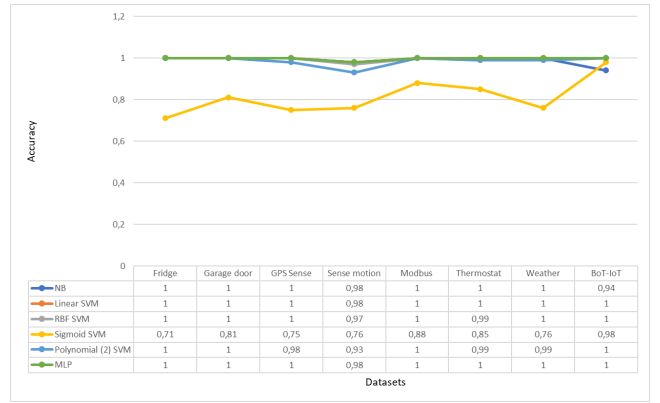


Fig. 4: Accuracy performance metric for the multi-class classification

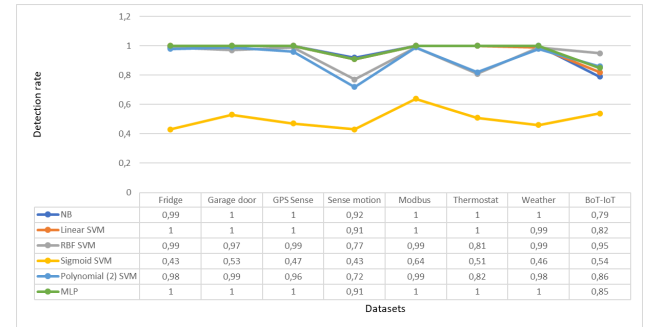


Fig. 5: Detection rate performance metric for the multi-class classification

Models	Traffic	Pre.	Re.	F1.
NB	DDos	0.93	0.96	0.94
	Dos	0.96	0.91	0.93
	Reconnaissance	0.98	0.78	0.86
	Theft	1	0.36	0.53
	Normal	0.99	0.98	0.99
Linear SVM	DDos	1	1	1
	Dos	1	1	1
	Reconnaissance	0.98	0.76	0.85
	Theft	1	0.36	0.53
	Normal	1	1	1
RBF SVM	DDos	1	1	1
	Dos	1	1	1
	Reconnaissance	0.99	0.85	0.91
	Theft	1	0.93	0.96
	Normal	1	1	1
Sigmoid SVM	DDos	0.98	0.99	0.98
	Dos	1	0.99	0.99
	Reconnaissance	0	0	0
	Theft	0	0	0
	Normal	0.75	0.74	0.75
Polynomial SVM	DDos	1	1	1
	Dos	1	1	1
	Reconnaissance	1	0.96	0.98
	Theft	0.83	0.36	0.5
	Normal	1	1	1
MLP	DDos	1	1	1
	Dos	1	1	1
	Reconnaissance	1	0.93	0.97
	Theft	1	0.36	0.53
	Normal	1	1	1

TABLE VI: Precision, recall and F1-score performance metrics with the multi-class classification for the BoT-IoT dataset

Class	NB			Linear SVM			RBF SVM			Sigmoid SVM			Polynomial SVM			MLP		
	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.	Pre.	Re.	F1.
Backdoor	1	1	1	1	1	1	1	1	1	0.17	0.38	0.23	1	0.99	0.99	1	1	1
DDoS	1	1	1	1	1	1	1	1	1	0.39	0.34	0.37	1	1	1	1	1	1
Injection	1	1	1	1	1	1	1	1	1	0.65	0.49	0.56	1	1	1	1	1	1
Password	1	1	1	1	1	1	1	1	1	0.41	0.31	0.36	0.99	1	1	1	1	1
Ransom	1	1	1	1	1	1	0.97	1	0.99	0.45	0.39	0.42	0.97	0.95	0.96	1	1	1
Scanning	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
XSS	1	1	1	1	1	1	1	0.96	0.98	0.33	0.21	0.26	0.93	0.96	0.94	1	1	1
Normal	1	1	1	1	1	1	1	1	1	1	0.95	0.98	1	1	1	1	1	1

(a) Performance metrics with the multi-class classification for the Fridge dataset

Backdoor	1	1	1	1	1	1	0.99	0.99	0.45	0.63	0.52	1	0.99	0.99	1	1	1	1
DDoS	1	1	1	1	1	1	0.99	0.99	0.5	0.19	0.28	1	1	1	1	1	1	1
Injection	1	1	1	1	1	1	0.99	1	0.99	0.8	0.96	0.87	0.99	1	0.99	1	1	1
Password	1	1	1	1	1	1	0.99	0.99	0.99	0.38	0.53	0.44	1	1	1	1	1	1
Ransom	1	1	1	1	1	1	0.95	1	0.98	0.71	0.65	0.68	0.98	1	0.99	1	1	1
Scanning	1	1	1	1	1	1	1	0.99	1	0.18	0.02	0.04	1	1	1	1	1	1
XSS	1	1	1	1	1	1	1	0.88	0.94	0.3	0.16	0.2	1	0.96	0.98	1	1	1
Normal	1	1	1	1	1	1	1	1	1	1	0.99	1	1	1	1	1	1	1

(b) Performance metrics with the multi-class classification for the Garage Door dataset

Backdoor	1	1	1	1	1	1	1	1	1	0.67	0.77	0.72	1	1	1	1	1	1
DDoS	1	1	1	1	1	1	1	1	1	0.24	0.22	0.23	0.95	0.83	0.88	1	1	1
Injection	1	1	1	1	1	1	1	1	1	0.67	0.48	0.56	1	1	1	1	1	1
Password	1	1	1	1	1	1	1	1	1	0.41	0.49	0.45	0.84	0.95	0.9	1	1	1
Ransom	1	1	1	1	1	1	0.99	1	0.99	0.84	0.76	0.8	0.99	1	0.99	1	1	1
Scanning	1	1	1	1	1	1	1	1	1	0	0	0	1	1	1	1	1	1
XSS	1	1	1	1	1	1	1	0.94	0.97	0.14	0.14	0.14	1	0.94	0.97	1	1	1
Normal	1	1	1	1	1	1	1	1	1	0.9	0.92	0.91	1	1	1	1	1	1

(c) Performance metrics with the multi-class classification for the GPS dataset

Backdoor	1	1	1	1	1	1	1	1	1	0.4	0.64	0.49	1	1	1	1	1	1
DDoS	1	1	1	1	1	1	0.99	1	1	0.25	0.3	0.27	1	0.59	0.74	1	1	1
Injection	1	1	1	1	1	1	1	1	1	0.74	0.77	0.75	1	1	1	1	1	1
Password	0.82	0.93	0.87	1	1	1	0.77	0.99	0.87	0.31	0.19	0.23	0.58	0.99	0.73	0.8	1	0.89
Ransom	1	1	1	0.8	1	0.89	0.85	1	0.92	0.63	0.58	0.6	0.85	1	0.92	1	1	1
Scanning	0.69	0.46	0.55	1	0.31	0.48	0.94	0.23	0.37	0.15	0.07	0.09	0.86	0.21	0.34	1	0.32	0.48
XSS	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	1	1	1
Normal	1	1	1	1	1	1	1	1	1	1	0.98	0.99	1	1	1	1	1	1

(d) Performance metrics with the multi-class classification for the Sense Motion dataset

Backdoor	1	1	1	1	1	1	1	1	1	0.58	0.71	0.64	1	1	1	1	1	1
DDoS	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Injection	1	1	1	1	1	1	1	1	1	0.59	0.66	0.62	1	1	1	1	1	1
Password	1	1	1	1	1	1	1	1	1	0.9	0.9	0.9	1	1	1	1	1	1
Ransom	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Scanning	1	1	1	1	1	1	0.96	0.98	0.97	0.42	0.29	0.34	0.98	1	0.99	1	1	1
XSS	1	1	1	1	1	1	1	0.98	0.99	0.56	0.34	0.42	1	1	1	1	1	1
Normal	1	1	1	1	1	1	1	1	1	0.99	0.96	0.98	1	1	1	1	1	1

(e) Performance metrics with the multi-class classification for the Modbus dataset

Backdoor	1	1	1	1	1	1	1	1	1	0.45	0.55	0.52	1	1	1	1	1	1
DDoS	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Injection	1	1	1	1	1	1	1	1	1	0.69	0.86	0.77	1	1	1	1	1	1
Password	1	1	1	1	1	1	1	1	1	0.57	0.41	0.47	1	1	1	1	1	1
Ransom	1	1	1	1	1	1	0.85	1	0.92	0.75	0.69	0.72	0.85	1	0.92	1	1	1
Scanning	1	1	1	1	1	1	1	0.73	0.85	0	0	0	1	0.8	0.89	1	1	1
XSS	1	1	1	1	1	1	0	0	0	0.12	0.11	0.11	0	0	0	1	1	1
Normal	1	1	1	1	1	1	1	1	1	0.98	0.97	0.97	1	1	1	1	1	1

(f) Performance metrics with the multi-class classification for the Thermostat dataset

Backdoor	1	1	1	1	1	1	1	1	1	0.33	0.44	0.38	1	1	1	1	1	1
DDoS	1	1	1	1	1	1	1	1	1	0.43	0.61	0.51	0.97	0.92	0.94	1	1	1
Injection	1	1	1	1	1	1	1	1	1	0.4	0.46	0.43	1	1	1	1	1	1
Password	1	1	1	1	1	1	1	1	1	0.88	0.56	0.68	0.93	0.97	0.95	1	1	1
Ransom	1	1	1	1	1	1	0.99	1	1	0.73	0.67	0.7	0.99	1	1	1	1	1
Scanning	1	1	1	1	1	1	1	1	1	0.01	0.01	0.01	1	1	1	1	1	1
XSS	1	1	1	1	0.99	1	1	0.98	0.99	0.23	0.06	0.1	1	0.98	0.99	1	1	1
Normal	1	1	1	1	1	1	1	1	1	0.98	0.93	0.96	1	1	1	1	1	1

(g) Performance metrics with the multi-class classification for the Weather dataset

TABLE VII: Performance metrics with the multi-class classification for the TON-IoT datasets

trained our proposed supervised ML models (i.e., NB, Linear SVM, RBF SVM, Sigmoid SVM, Polynomial SVM and MLP) based on these datasets.

In the state of the art, ML approaches are divided into three approaches : Similarity-based, Probability-based and Boundary decision-based [40]. In our investigation, we select the best-performing algorithms in recent studies due to several reasons. Indeed, NB has demonstrated, in related works, its effectiveness when dealing with huge datasets and a small number of features [41]; SVM ranks top of the list of different classifiers in the case of small and medium datasets; and MLP, as a feed-forward model, has proven its ability to model the features space to detect attacks with non-linear approximation throughout the weights back-propagation learning process. It may extract a sophisticated relationship between variables to

learn an IoT attack.

This study advanced our understanding of the effectiveness of ML methods for anomaly detection of the intrusion detection system in the IoT networks. It was found that for binary classification detection, the NB, Linear SVM, RBF SVM, polynomial SVM, and the MLP models show prominent results for the different performance metrics (i.e., accuracy, detection rate, precision, recall, and F1-score). However, for the multi-class classification detection, the Linear SVM and the MLP show better performance metrics for the different studied datasets. These models were also able to provide good performance for less frequent attacks, which has rare samples in the datasets. On the other hand, the sigmoid SVM model shows the lowest performance metrics, and it was ineffective in detecting infrequent attacks.

We notice that the linear SVM gives more performed results than the other SVM model kernels (i.e., RBF, Polynomial and Sigmoid). Indeed, it is clear from Table II and Table III that the datasets have imbalanced categories of attacks. In such a case, a non-linear SVM could face issues in classifying the different categories of attacks. MLP, through its perceptron considering a linear separator in a high dimensional space, resolves the lack of data in some categories of attacks and depends only on enough training samples, which is the case of our datasets.

Our results suggest that there are models that can address the issues of unbalanced datasets and provide satisfactory detection performance in IoT network context; the Linear SVM and MLP models can be such an example.

IV. CONCLUSION

In this work, we aim to investigate the ability to detect attacks that can compromise the CIA Triad in addition to detecting the less frequent attacks. Therefore, we have studied two relevant datasets for IDS anomalies detection for IoT networks (i.e., Bot-IoT and TON-IoT). The TON-IoT datasets are composed of Fridge, Garage door, GPS, Sense motion, Modbus, Thermostat, and Weather datasets. Indeed, we explored their simulated IoT services, considered cyber-attack scenarios, and different features. In addition, we dressed a comparison between them by showing their common and distinguished characteristics. Thereafter, we have proposed different models for IDS anomaly detection using a bunch of ML methods (i.e., NB, Linear SVM, RBF SVM, Sigmoid SVM, Polynomial (2) SVM, and MLP). Moreover, we studied the performance of these models using accuracy, detection rate, recall, precision, and F1-score performance metrics. The results show that the NB, Linear SVM, RBF SVM, polynomial SVM, and MLP models provide interesting result with binary classification. However, the Linear SVM and MLP model outperform the other models with the multi-class classification.

This work will be extended to study active and deep learning methods to enhance detection performance in unbalanced dataset, in addition to robust methods for features selection.

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